{DALAS} Drug Repurposing Project

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The goal of this project is to identify promising candidate drug-disease associations for new clinical trials by analyzing already approved clinical indications and research data. To this end, we have collected three types of data - about drugs, about diseases and about associations between them, restricting our analysis to the well-defined heterogeneous but mechanistically related group of autoimmune diseases.

Dataset	**Source**	**#Records**	**#Features**	**IDs**
Drugs	ChEMBL	679	43	ChEMBL / UniProt
Diseases	Open Targets	42	7	EFO / MeSH
Associations	ClinicalTrials.gov	1314 trials for 1256 associations	9	NCT

First, we scraped 50 disease MeSH IDs from the page of NIH (National Institute of Health) with classification of autoimmune diseases. We then used those identifiers to get the chemical data about 679 drugs that are indicated for 42 of those diseases (or are currently under investigation) from the ChEMBL database via the ChEMBL Python client.

Among these chemical data are some basic physico-chemical and structural properties, number of aromatic rings, count of heavy atoms, molecular weight, chirality, optimal pH, etc., as well as string representations of their actual chemical structure that we used to extract numerical features (molecular fingerprints) using RDKit package. Using the same ChEMBL database, we have managed to extract a list of protein targets for some of those drugs with the association e.g. inhibitor / activator. We had to standardize corresponding UniProtIDs and get rid of synonyms using the UniProt python ID mapper. In the end we obtained 43 drug descriptors. We may need to split this dataset in two to avoid a lot of missing data because of the nature of some drugs - big biomolecules, e.g. antibodies, and small molecules don't have exactly the same molecular characteristics (see annexe for missingness heatmap and correlation heatmap on a subset of numerical features and binary flags). Other missing chemical data is either to be replaced with the appropriate default value or imputed, e.g. with kNN, and further validated by comparing imputations on subsets with known values. Among identified drugs there are some outliers corresponding to rare drug types such as inorganic molecules, often missing key descriptors and having unusual physicochemical profiles (see annexe). As for targets, only two thirds of drugs have at least one identified protein target, probably, we'd have to extend the scope of considered targets to DNA / RNA sequences to avoid omitted variable / selection bias.

Next, we extracted disease data by sending GraphQL requests to Open Targets API. To do this, we had to map disease's MeSH IDs to EFO IDs, using cross-references provided by ChEMBL. We retrieved following data: medical textual description,vectors of associated phenotypes (symptoms and their textual descriptions), protein targets (molecules involved in pathogenesis) as well as evidence of the corresponding target association, e.g. hyper or hypofunction of the target - only 7 descriptors overall. Data is mostly complete except for phenotypes which are present only for 14 out of 42 diseases. We'll have to either exclude this variable or try to complete it with other databases. As for targets, each disease has at least 5 associated targets and a rich body of evidence for those associations, although more work is required to standardize those associations (e.g. infer hyperfunction if blocking this target improves disease outcomes). Also we had to map the ENSEMBL IDs of targets to the UniProt IDs using the same UniProt python ID mapper as for drug targets so that we can compare them.

Finally, to extract the data about drug-disease associations, we first addressed 1256 indications furnished by ChEMBL containing references (NCT IDs) to the ClinicalTrials.gov database that we connected via their REST API. We collected metadata of corresponding 1314 clinical trials, such as timestamps of start and end of the study, phase, its status (completed, ongoing, terminated), reason of termination if it's the case (textual explanation e.g. drug is not efficient, toxic, or lack of money to finish the study), as well as results if such are present. About two thirds of the studies are in phase II or III, two thirds are completed, and only one third of studies has results. It's important to notice that there's no direct indication to whether the trial was successful or not, so we had to figure it out ourselves from the results. First, those drug-disease indications that are referenced as already approved (referenced by DailyMed medical recommendations in ChEMBL) or that reached phase IV (155 associations) are automatically considered as successful. Others are supposed to be unknown unless their inefficacy was shown. The tricky part was to figure out whether it was the case. In order to do so, we extracted p-values from results and considered the trial as failed if the majority of its p-values are non-significant. It resulted in 50 studies marked as failed. We plan to probably extend this list by analysing textual reasons for termination of some studies manually or with language models to find trials that are terminated because of drug inefficiency. This overall success of drug-disease trials will serve as a label for model training ("success", "failure", "unknown"), although there's a potential label noise because of heuristics used.

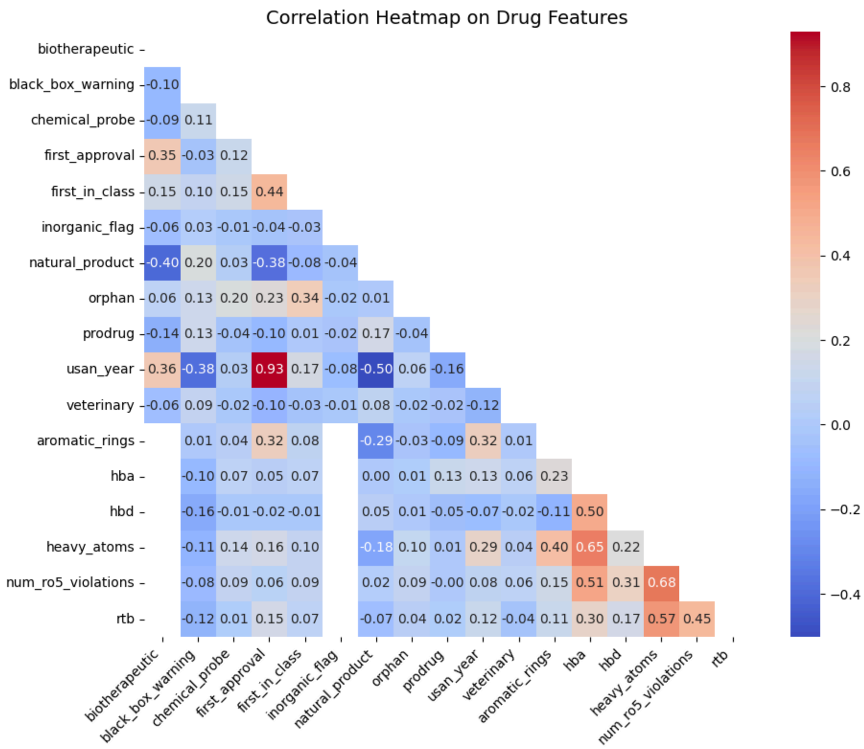
Also, we engineered some features for drug-disease associations (28518 combinations in total) such as proximity of their names in the embedding space of language models trained on biomedical scientific papers - we used SentenceTransformer and PubMedBERT from Hugging Face although we should try out different models. The interest of doing so is to see how associated corresponding drugs and diseases are in the scientific literature - in the research that usually precedes actual clinical trials by years or even decades. Of note, top proximities are assigned to drug-disease matches that are indeed approved medical indications. We would love to do the same thing with textual descriptions but drugs might be strongly biased towards the diseases they are primarily used against since usually text description of a drug contains a mention of primary use which might reduce the likelihood of discovering an alternative application. To handle this textual leakage issue, we can pre-treat drug textual descriptions and remove explicit disease mentions.

Other features that are a bit harder to engineer are based on shared targets between drugs and diseases revealing mechanistic similarities. We are still working on some challenges such as missing targets for some drugs, and the fact that we should take into account the direction of association with targets (positive / negative). As the last challenge, we have to map raw targets into lists of associated molecular Reactome pathways using the same UniProt python ID mapper since it makes more biological sense to compare pathways, but we still need to think how to handle directionality of associations and engineer more features based on pathway similarities.

Given the small number of known labels - “success” or “fail”, we will likely extend the scope of the project from autoimmune diseases to the disorders of the immune system in general to be able to train a more robust model. As for drug-disease pairs with “unknown” labels, we would like to use them for unsupervised models - clustering and use assigned clusters as a feature for supervised learning on data with known labels - e.g. logistic regression, random forest, XGBoost using techniques to compensate for imbalanced label distribution such as class weighting or synthetic oversampling. The supervised model aims to predict likelihood of success of a given drug-disease trial, given its chemical, biological and semantic features. We can also leverage the intrinsic graph structure of our data - bipartite graph drugs-diseases connected with known indications and through shared pathways to apply link prediction algorithms, graph neural networks. The model can be evaluated by a cross-validation based on splitting disease matches by the timestamp of the first successful or failed trial, which simulates realistic prediction situations - prediction of new successful studies based on old ones. The ultimate scientific interest is to apply the final classification model to "unknown" drug-disease associations and sort them by the success potential, although, for obvious reasons, there’s no ground truth available.

It’s also important to highlight that some drug-disease combinations might have not been tried for a reason - they might be actually some hidden “premature failures” that have not been tried because of some incompatibility obvious to experts, we can try to handle this issue by finding criteria for preliminary exclusion of most irrelevant combinations. This issue is exacerbated by the publication bias - successful trials are more likely to be reported in the literature, leaving such “premature failures” unpublished. In some cases, the data missingness depended on the diseases in question - some diseases such as diabetes, rheumatoid arthritis are better annotated than others - database curation bias. Finally, it’s important to note one of most obvious discrimination biases in the case of analysing biomedical data - is that databases are largely biased towards western populations (USA, Europe…) and thus towards biological and molecular traits characteristic of white populations (e.g. protein target variants, adverse effects of drugs…).

ANNEXE


Correlation Heatmap on Drug Features
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black_box_warning
chemical_probe
first_approval
first_in_class
inorganic_flag
natural_product
orphan
prodrug
usan_year
veterinary
aromatic_rings
hba
hbd
heavy_atoms
num_ro5_violations
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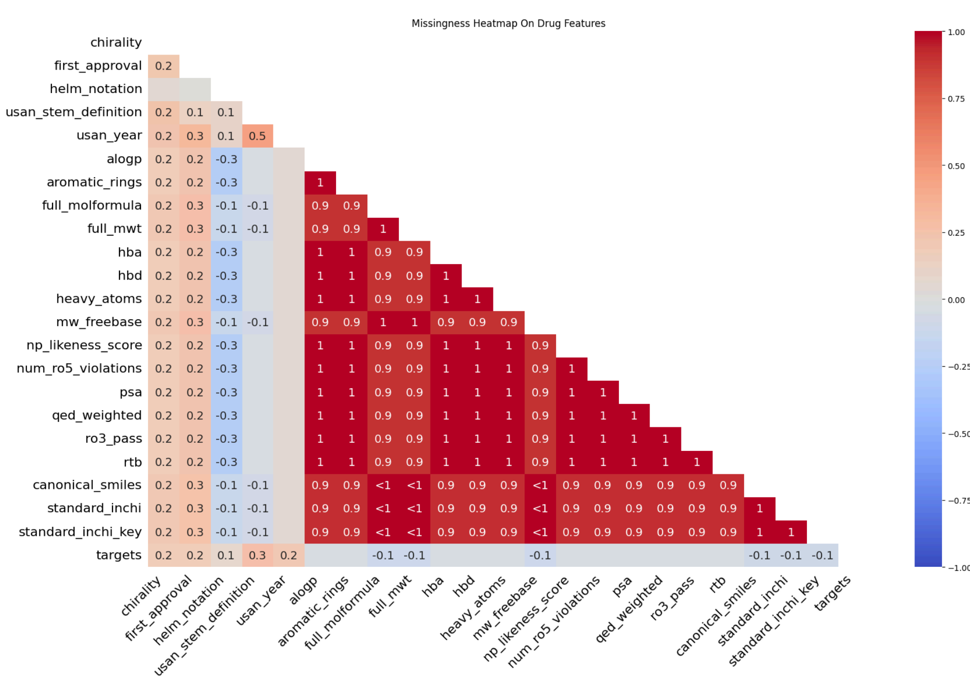
0.4
0.2
0.0
−0.2
−0.4


Missingness Heatmap On Drug Features
chirality
first_approval
helm_notation
usan_stem_definition
usan_year
alogp
aromatic_rings
full_molformula
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hbd
heavy_atoms
mw_freebase
np_likeness_score
num_ro5_violations
psa
qed_weighted
ro3_pass
rtb
canonical_smiles
standard_inchi
standard_inchi_key
targets